p-adic Apéry limits: rate laws, valuation strata, and closed forms via the p-adic Gamma function

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Draft, July 25, 2026

Abstract

Let $(A(n), B(n))$ be an Apéry-like pair: A one of the fifteen sporadic binomial sums, B the second solution of its recurrence. Classically $B(n)/A(n)$ converges to the *Apéry limit*, a rational multiple of $\zeta(w)$ or of $L(\chi, w)$, where w is the weight and χ a quadratic character attached to the sequence. We study the p-adic analogue. Along a p-power tower $n_s = ap^s$ the normalised ratios $\chi(p)^s p^{ws} B(n_s)/A(n_s)$ converge in $\mathbb{Q}_p$; we call the limit Λ_a a *p-adic Apéry limit*. Convergence is driven by the digit-descent congruences $v_p\big(p^w B_n A_q - \chi(p) B_q A_n\big) \geq w$, $q = \lfloor n/p \rfloor$, of the companion Paper A.

We report four results. **(1) A rate law.** The number of new certified p-adic digits per tower level is $\min\big(3,\, r(\chi, w)\big)$, where the 3 is the Jacobsthal–Kazandzidis exponent governing $\binom{Ap^t}{Bp^t}$ and $r(\chi, w)$ is the rate of the χ-twisted harmonic tower. The rate is therefore *not* the motivic weight, correcting a natural guess; for weight 5 the only correct multi-digit statement is the scaled congruence $p^{5s} P_{ap^s} Q_{ap^{s-1}} \equiv p^{5(s-1)} P_{ap^{s-1}} Q_{ap^s} \pmod{p^{3s}}$, of depth $3s$. **(2) Valuation strata and a vanishing defect.** For the Brown–Zudilin $\zeta(5)$ family, $v_p(\widehat{P}_n/Q_n) = -3L + O(1)$ and $v_p(P_n/Q_n) = -5L + O(1)$ with $L = \lfloor \log_p n \rfloor$ and the $O(1)$ bounded by 2; consequently the p-adic purity defect vanishes identically, $c_p = \lim_n \widehat{I}_n/I'_n = 0$, in sharp contrast with the archimedean defect $c = -3/\pi^2$. **(3) An exclusion theorem.** No tower limit of the classical Apéry pair or of the Brown–Zudilin pair is a $\mathbb{Q}$-affine function of $\zeta_p(3)$ or $\zeta_p(5)$ of height $\lesssim 10^4$; structurally, the p-adic linear forms in $\zeta_p(3)$, $\zeta_p(5)$ built from these pairs *diverge polynomially*, so no sharpening can rescue them. These pairs do not p-adically approximate the Kubota–Leopoldt zeta values at all. **(4) An identification that explains (3).** We exhibit, and verify block by block, a decomposition $\Lambda_a = h_p(a) + (\text{blocks})/\widetilde{A}_a$ with $h_p(a) = \sum_{x \in \mathbb{Z}[1/p],\, 0<x\leq a} x^{-3}$, every block a polynomial in the Kazandzidis limits $c_p(A, B) = \lim_t \binom{Ap^t}{Bp^t}$; and we *prove* the two ingredients of that decomposition — the closed form of h_p, and $c_p(A, B)$ in closed form: -1 times a p-power times a ratio of Morita Γ_p-values times exp of an explicit $\mathbb{Q}$-series in *all* odd p-adic zeta values. Thus Λ is a *value* of the Beukers–Vlasenko generating function $\Gamma_p(x) e^{-\Gamma'_p(0)x} = \exp\big(-\sum_{m\geq 2} \zeta_p(m) x^m/m\big)$ at p-power arguments, whereas the Beukers–Vlasenko Frobenius structure constants are its Taylor coefficients at 0 — and for the order-3 Apéry operator those slots are literally zero.

Finally, three of the fifteen families have no archimedean Apéry limit at all — their characteristic roots are complex conjugates of equal modulus — and yet they do have p-adic tower limits.

Contents

1 Introduction

1.1 Apéry limits, classically

Apéry's irrationality proof for $\zeta(3)$ [4, 14] produces two solutions of one three-term recurrence,

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5)\, u_n - n^3 u_{n-1}, \tag{1.1}$$

namely the integers

$$a_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$$

and the second solution b_n normalised by $b_0 = 0$, $b_1 = 6$; the pair satisfies $b_n/a_n \to \zeta(3)$. The number $\zeta(3)$ appears here not as an integral but as a *limit of a ratio of two solutions of a difference equation*. This is the phenomenon that has come to be called an *Apéry limit*.

It is not isolated. Zagier's search for integral solutions of the $(n+1)^2$-recurrence [33], the Almkvist–Zudilin list at $(n+1)^3$, and Cooper's three additional cases [13] together give fifteen *sporadic* Apéry-like pairs. We use throughout the labelling and data of Malik–Straub [26] and Straub [31], reproduced in Table 1. Each pair has a weight $w \in \{2, 3\}$ — the minimal exponent with $d_n^w B(n) \in \mathbb{Z}$, $d_n = \operatorname{lcm}(1, \dots, n)$ — and an Apéry limit which is a rational multiple of $\zeta(w)$ or of $L(\chi_D, w)$ for a quadratic character χ_D. The character of the limit turns out to be an invariant of the whole p-adic story as well; see Finding 7.2.

Beyond the sporadic list, the same shape occurs in the Brown–Zudilin cellular approximations to $\zeta(5)$ [9], whose conceptual setting — Apéry-type proofs as cell integrals on moduli spaces, and the structural reason $\zeta(5)$ resists — is Brown's [10]. There one has a rank-3 system with rows Q_n, $\widehat{P}_n$, P_n and *two* Apéry limits: $\widehat{P}_n/Q_n \to \zeta(3)$ and $P_n/Q_n \to \zeta(5)$. This is our weight-5 test case throughout.

1.2 What is known p-adically

Almost nothing, and in particular nothing named. The literature contains an extensive study of the *convergence* of the towers we are about to use — the supercongruences $A(mp^r) \equiv A(mp^{r-1}) \pmod{p^{3r}}$ of Beukers, Coster and Ishikawa and their descendants [32], and the Dwork-crystal framework of Beukers–Vlasenko [8] which explains the A-row congruences as the unit-root entry of a Frobenius matrix — but the *limits* of the corresponding second-solution towers do not appear to have been studied at all. Our own searches turned up no name, no notation and no closed form for them; see §6.5 for what we did and did not find.

There are two nearby bodies of work, and it is important to say precisely how our object differs from each.

- *p-adic zeta values and irrationality.* Calegari [11] proved $\zeta_p(3) \notin \mathbb{Q}$ for $p = 2, 3$, and Lai–Sprang–Zudilin [25] proved $\zeta_2(5) \notin \mathbb{Q}$; the asymptotic p-adic Ball–Rivoal theory is developed in [24]. A recurring expectation — "one ratio, two completions" — is that the Apéry pair should approximate $\zeta_p(3)$ as it approximates $\zeta(3)$. Section 5 shows this is *false*, and false for a structural reason. Consistently, Calegari's proof does not use the Apéry pair at all.
- *Frobenius constants.* There is a genuine terminological collision here. The Bloch–Vlasenko / Roy–Vlasenko *Frobenius constants* κ_n [29, 20] are archimedean periods. The Beukers–Vlasenko *Frobenius structure constants* α_j [7] are p-adic, and are Taylor coefficients of $\Gamma_p(x)e^{-\Gamma'_p(0)x}$ at $x = 0$. Our Λ_a is the p-adic analogue of neither: it is the analogue of the *Apéry limit itself*, i.e. of a κ-type period. It happens to be built from the same generating function as the α_j, but by *evaluation* at p-power arguments rather than by Taylor expansion at 0. See §6.5.

1.3 Results

Fix an odd prime $p \geq 5$ and a tower base a with $p \nmid a$, and set $n_s = ap^s$.

(A) Existence and the rate law (§2–§3). The digit-descent congruences of Paper A [1],

$$(\mathrm{LB}^{\chi}_{w}) \qquad v_p\big(p^w B_n A_q - \chi(p) B_q A_n\big) \;\geq\; w, \qquad q = \lfloor n/p \rfloor, \qquad p \geq 5,$$

make the sequence $L_s := \chi(p)^s p^{ws} B(n_s)/A(n_s)$ Cauchy, so that

$$\Lambda_a \;:=\; \lim_{s\to\infty} \chi(p)^s\, p^{ws}\, \frac{B(ap^s)}{A(ap^s)} \;\in\; \mathbb{Q}_p$$

exists (Theorem 2.4). The limit is an intrinsic invariant of the recurrence and the tower: it is unchanged by $B \mapsto B + cA$, scales by λ under $B \mapsto \lambda B$, and satisfies $\Lambda_{p^k a} = \chi(p)^k p^{-wk} \Lambda_a$ (Proposition 2.5). The observed convergence rate is

$$\boxed{\text{digits per level} \;=\; \min\big(3,\; r(\chi, w)\big)}$$

where $r(\chi, w)$ is the rate of the χ-twisted harmonic tower (Table 2): $r(\chi, w) = w$ for $\chi \neq 1$, and $r(1, w) = w + 1$, improved to $w + 2$ for odd w because $\zeta_p(w + 1) = 0$. The 3 is *Jacobsthal–Kazandzidis*, not the motivic weight (Finding 3.3). A first reading of the weight-3 and weight-5 data — "the rate is always exactly 3" — is an artefact of testing only untwisted pairs; the χ-twisted weight-2 families converge at rate 2.

(B) The weight-5 correction (§3.4). For the Brown–Zudilin P-row the normalising power is p^{5s} — $v_p(p^{5s} P_{n_s}/Q_{n_s})$ is constant in s — but the single-step ratio descent $p^5 f(n_s) \equiv f(n_{s-1})$ has depth $5 - 2s$, which goes negative at $s \geq 3$. The surviving multi-digit law is the scaled one,

$$p^{5s}\, P_{ap^s}\, Q_{ap^{s-1}} \;\equiv\; p^{5(s-1)}\, P_{ap^{s-1}}\, Q_{ap^s} \pmod{p^{3s}},$$

of depth exactly $3s$ (Finding 3.9).

(C) Strata and the vanishing defect (§4). With $L = \lfloor \log_p n \rfloor$ and over *all* n, not merely towers,

$$v_p(Q_n) \geq 0, \qquad v_p(\widehat{P}_n/Q_n) = -3L + O(1), \qquad v_p(P_n/Q_n) = -5L + O(1),$$

the $O(1)$ lying in $[-2, 1]$ uniformly (Finding 4.1). Hence the two Brown–Zudilin linear forms $\widehat{I}_n = Q_n\zeta(3) - \widehat{P}_n$ and $I'_n = Q_n\zeta(5) - P_n$ sit in different valuation strata, $v_p(\widehat{I}_n/I'_n) = 2L + O(1) \to +\infty$, and the p-adic purity defect vanishes:

$$c_p := \lim_{n\to\infty} \widehat{I}_n/I'_n = 0 \qquad \text{in } \mathbb{Q}_p, \quad p \in \{5, 7, 11, 13, 17, 19\}$$

(Finding 4.3). Archimedeanly the same limit is $c = -1/(2\zeta(2)) = -3/\pi^2 \neq 0$, and that non-vanishing is exactly what exiles the Brown–Zudilin class from the minimal ray [2]. The contrast is structural and is discussed in §4.3: the archimedean side is obstructed by *positivity*, the p-adic side is unobstructed because the Frobenius *grading* $\mathrm{diag}(1, p^3, p^5)$ separates the rows.

(D) The exclusion theorem (§5). This is a headline negative and we state it as one. Using an implementation of the Kubota–Leopoldt p-adic L-function built from scratch and validated on 980/980 interpolation identities (Appendix A):

- No relation $c_0 + c_1\zeta_p(w) + c_2\Lambda = 0$ of height $\lesssim 10^4$ holds, for Λ any of Λ_a (Apéry), Λ_a^P, $\Lambda_a^{\widehat{P}}$ (Brown–Zudilin), at $p = 5, 7, 11, 13$, to 12–27 certified p-adic digits (Finding 5.1). Every candidate returned by exact LLL sits *at* the noise floor.
- The sharp form of the negative is not about margins. For all $n \leq 5000$ and $p \in \{5, 7, 11, 13\}$,

  $$v_p\big(b_n - \zeta_p(3)a_n\big) - v_p(a_n) = -3\lfloor \log_p n \rfloor + O(1),$$
  $$v_p\big(Q_n\zeta_p(5) - P_n\big) - v_p(Q_n) = -5\lfloor \log_p n \rfloor + O(1)$$

  (Finding 5.3). The p-adic linear forms *diverge polynomially* where a Bel-type irrationality criterion [25, Lemma 4] needs $\exp(-\alpha n)$. The gap is infinite, not finite.
- Structurally: the Apéry operator $\theta^3 - t(34\theta^3 + 51\theta^2 + 27\theta + 5) + t^2(\theta + 1)^3$ has order 3, so in the Beukers–Vlasenko Frobenius expansion only α_1, α_2 exist — and by [7, Thm. 1.5] both are 0. The value $\zeta_p(3)$ first occupies α_3, which requires order ≥ 4 (§6.5).

(E) The identification (§6). The exclusions are then *explained*, not merely observed. Apéry's formula splits b_n/a_n into a harmonic part and a remainder, and both parts have exact p-adic limits:

$$h_p(a) := \lim_s p^{3s} H_3(ap^s) = \sum_{x \in \mathbb{Z}[1/p],\ 0 < x \leq a} x^{-3} \qquad \text{(Theorem 6.1)},$$

$$c_p(A, B) := \lim_{t\to\infty} \binom{Ap^t}{Bp^t} = -p^{v_p\binom{A}{B}}\, T(A, B)\, \frac{\Gamma_p(A+1)}{\Gamma_p(B+1)\Gamma_p(A-B+1)}\, e^{\mathcal{Z}(A,B)} \qquad \text{(Theorem 6.3)},$$

where T is an explicit finite Γ_p-product over the base-p truncations $\lfloor A/p^l \rfloor$, $\lfloor B/p^l \rfloor$, $\lfloor (A-B)/p^l \rfloor$ $(l \geq 1)$, and

$$\mathcal{Z}(A, B) = -\sum_{\substack{m \geq 3 \\ m \text{ odd}}} \zeta_p(m)\left(A^m - B^m - (A-B)^m\right)\frac{p^m}{m(1-p^m)}.$$

Assembling, $\Lambda_a = h_p(a) + (\text{convergent block sum})/\widetilde{A}_a$ with every block a polynomial in the $c_p(\cdot,\cdot)$ (Finding 6.6). Since each $c_p(A,B)$ is the exponential of an infinite $\mathbb{Q}$-combination of *all* of $\zeta_p(3), \zeta_p(5), \zeta_p(7), \ldots,$ no finite low-height $\mathbb{Q}$-relation with $\zeta_p(3)$ alone can exist. The quantitative form of this: truncating the series after $\zeta_p(3)$ reproduces 4–5 digits out of 24–30 [VERIFIED: $p = 5, 7, 11, 13$]. The same mechanism refutes the hypothesis that the family $\{\Lambda_a\}$ is generated over $\mathbb{Q}$ by a single constant (§6.6).

(F) Existence without archimedean limits (§7). Three of the fifteen families — B, (δ), (η) — have characteristic roots that are complex conjugates of equal modulus, so $B(n)/A(n)$ oscillates forever and there is *no* archimedean Apéry limit. All three nevertheless have p-adic tower limits: (δ) outright at rate 3, and B, (η) after the $\chi(p)^{-s}$ twist. The p-adic Apéry limit exists where the archimedean one does not.

1.4 Method, and how to read the evidence tags

Statements in this paper carry an explicit evidence class, printed with the statement.

[PROVED]
: A complete proof is given here or in the cited companion paper.

[VERIFIED: range]
: A finite exact computation over the stated range, at the stated p-adic precision. This is evidence, never proof, and the range is always stated.

[RECALLED–UNVERIFIED]
: Asserted from memory of the literature and not checked against a source in the preparation of this paper.

[CONJECTURAL FIT]
: A numerical fit reproducing known data points, extrapolated.

Statements labelled FINDING are supported by [VERIFIED:] evidence only; statements labelled THEOREM, PROPOSITION, LEMMA, COROLLARY are [PROVED]. We have been deliberate about not promoting one class to the other: several of the most interesting statements below (the rate law, the strata bounds, $c_p = 0$) are at present Findings.

Remark 1.1 (Methodology). All numerical statements below were produced by exact integer and rational arithmetic — no floating point occurs anywhere in the pipeline, and every p-adic digit is obtained from an exact rational — and each closed form recorded here was checked against an independent direct computation of the object it describes, written separately from the code that produced the form. Ranges, certified precisions and the number of independent checks are stated with each claim; the programs are listed in Appendix D.

1.5 Notation and conventions

p denotes an odd prime, always ≥ 5 unless stated otherwise. We write $q_p = p$ (and $q_2 = 4$), ω for the Teichmüller character modulo q_p, and $\langle x \rangle = x/\omega(x)$. Following [25, Def. 8] we set

$$\zeta_p(s) := L_p\big(s, \omega^{1-s}\big) \qquad (s \in \mathbb{Z},\ s \geq 2),$$

so $\zeta_p(3) = L_p(3, \omega^{-2})$ and $\zeta_p(5) = L_p(5, \omega^{-4})$; this is the same normalisation as [7], for which $\zeta_p(m)$ is the p-adic limit of $-(1-p^{n-1})B_n/n$ along $n = 1 - m + (p-1)p^r$. In particular $\zeta_p(2k) = 0$ for all $k \geq 1$, a vanishing that will be used repeatedly.

For the classical Apéry pair we write $f(n) = b_n/a_n$, and we compute with the integer normalisation $\alpha_n = (n!)^3 a_n$, $\beta_n = (n!)^3 b_n$, which satisfy

$$u_{n+1} = (34n^3 + 51n^2 + 27n + 5)u_n - n^6 u_{n-1}, \qquad (\alpha_0, \alpha_1) = (1, 5), \quad (\beta_0, \beta_1) = (0, 6), \quad (1.2)$$

so that $f = \beta/\alpha$. [VERIFIED: α_n against the closed-form double sum for $n \leq 29$, 0 mismatches; $f(400)$ against $\zeta(3)$ to 79 decimals; generated exactly to $n = 125\,000$, and mod p^K to $n = 400\,000$.]

For the Brown–Zudilin pair we use the printed normalisation of [9]: $Q_0 = 1$, $P_0 = 0$, $Q_1 = 21$, $P_1 = 87/4$, $P_2 = 1190161/384$, $\widehat{P}_1 = 101/4$, $\widehat{P}_2 = 344923/96$, and we set $I'_n = Q_n\zeta(5) - P_n$, $\widehat{I}_n = Q_n\zeta(3) - \widehat{P}_n$. [VERIFIED: the certified order-3 recurrence has residual exactly 0 on all three rows for every $n \leq 357$ (358 checks per row), and the exact extension reproduces $P_n/Q_n \to \zeta(5)$, $\widehat{P}_n/Q_n \to \zeta(3)$ to 59 decimals at $n = 800$ and $n = 5000$.]

Throughout $L = \lfloor \log_p n \rfloor$ and $q = \lfloor n/p \rfloor$.

1.6 Organisation

Section 2 recalls the descent congruences and turns them into a convergence engine. Section 3 is the rate law and its weight-5 correction. Section 4 is the strata theorem and $c_p = 0$. Section 5 is the exclusion theorem. Section 6 identifies the limits. Section 7 is the no-archimedean-limit trio. Section 8 lists open problems. Appendix A documents and validates the Kubota–Leopoldt implementation; Appendix B reproduces the Lai–Sprang–Zudilin denominator ledger; Appendix C collects numerical data; Appendix D lists the programs.

2 The descent congruences as the convergence engine

2.1 The sporadic families

Table 1 lists the fifteen sporadic Apéry-like pairs, in the labelling of [26]. The recurrences are

$$\text{(R2)} \quad (n+1)^2 u_{n+1} = (an^2 + an + b)u_n - cn^2 u_{n-1}, \qquad (2.1)$$

$$\text{(R3)} \quad (n+1)^3 u_{n+1} = (2n+1)(an^2 + an + b)u_n - n(cn^2 + d)u_{n-1}, \qquad (2.2)$$

with A the sporadic binomial sum and B the second solution normalised by $B(0) = 0$, $B(1) = 1$ (the recurrence being enforced for $n \geq 1$; the $n = 0$ step is degenerate — this is Apéry's own construction, and the normalisation of [12]). Rescaling B by a p-unit rational changes Λ_a only by that scalar, so for $p \geq 5$ this normalisation is canonical up to a harmless constant.

2.2 The twisted descent law

The input we take from Paper A [1] is the following universal digit-descent law. Write $q = \lfloor n/p \rfloor$.

Theorem 2.1 (Paper A [1])**.** [PROVED] *(single-digit form). Let $p \geq 5$, let (A, B) be a sporadic pair with weight w and character χ as in Table 1, and suppose B admits a χ-homogeneous*

Table 1: The fifteen sporadic pairs, their Apéry limits, weights and characters, after [26, 31]; limits identified in [1], several of them presumably also in the numerical tables of [3]. w is the minimal exponent with $d_n^w B(n) \in \mathbb{Z}$.

#	label	$A(n)$	Apéry limit	w	χ
1	A	$\sum_k \binom{n}{k}^3$ (Franel)	$\zeta(2)/4$	2	1
2	B	$\sum_k (-1)^k 3^{n-3k} \binom{n}{3k} \frac{(3k)!}{k!^3}$	*none*	2	χ_{-3}
3	C	$\sum_k \binom{n}{k}^2 \binom{2k}{k}$	$L_{-3}(2)/2$	2	χ_{-3}
4	D	$\sum_k \binom{n}{k}^2 \binom{n+k}{n}$ (Apéry $\zeta(2)$)	$\zeta(2)/5$	2	1
5	E	$\sum_k \binom{n}{k} \binom{2k}{k} \binom{2(n-k)}{n-k}$	$L_{-4}(2)/2 = G/2$	2	χ_{-4}
6	F	$\sum_k (-1)^k 8^{n-k} \binom{n}{k} \sum_l \binom{k}{l}^3$	$(5/8)L_{-3}(2)$	2	χ_{-3}
7	(α)	$\sum_k \binom{n}{k}^2 \binom{2k}{k} \binom{2(n-k)}{n-k}$ (Domb)	$7\zeta(3)/24$	3	1
8	(γ)	$\sum_k \binom{n}{k}^2 \binom{n+k}{n}^2$ (Apéry $\zeta(3)$)	$\zeta(3)/6$	3	1
9	(δ)	$\sum_k (-1)^k 3^{n-3k} \binom{n}{3k} \binom{n+k}{n} \frac{(3k)!}{k!^3}$	*none*	3	1
10	(ε)	$\sum_k \binom{n}{k}^2 \binom{2k}{n}^2$	$7\zeta(3)/32$	3	1
11	(ζ)	$\sum_{k,l} \binom{n}{k}^2 \binom{n}{l} \binom{k}{l} \binom{k+l}{n}$	$L_{-3}(3)/3$	3	χ_{-3}
12	(η)	$\sum_k (-1)^k \binom{n}{k}^3 \left[\binom{4n-5k-1}{3n} + \binom{4n-5k}{3n}\right]$	*none*	3	χ_5
13	s_7	$\sum_k \binom{n}{k}^2 \binom{n+k}{k} \binom{2k}{n}$	$\zeta(2)/7$	2	1
14	s_{10}	$\sum_k \binom{n}{k}^4$	$\zeta(2)/5$	2	1
15	s_{18}	Cooper's s_{18}	$L_{-3}(2)/2$	2	χ_{-3}

harmonic-monomial decomposition of weight w on the summand of A satisfying the tameness and Lucas hypotheses of [1]. Then for $1 \le a < p$, $0 \le r < p$ and $n = ap + r$,

$$p^w B(ap+r) \;\equiv\; \chi(p)^e B(a) A(r) \pmod{p},$$

with $e \in \{0,1\}$ the number of χ-letters in each monomial. The hypotheses are verified there for (γ), A, s_{10} ($p \ge 5$, resp. $p \ge 7$) and D ($p \ge 7$).

Finding 2.2 (the master form, Paper A)**.** [VERIFIED: all 15 sporadic pairs; 28 primes $5 \le p \le 103$; $n \le 400$ for all p and $n \le 1200$ for $p = 5, 7, 11$; exact rational arithmetic; zero failures]

$$(\mathrm{LB}_w^\chi) \qquad v_p\big(p^w B_n A_q - \chi(p) B_q A_n\big) \;\ge\; w, \qquad q = \lfloor n/p \rfloor, \qquad p \ge 5,$$

and the floor is *exactly* w in every single (sequence, prime) cell: no κ-dependence, no centre-residue exception, no exceptional-digit exception. For the classical Apéry pair this reads $p^3 b_n a_q \equiv b_q a_n \pmod{p^3}$, [VERIFIED: $5 \le p \le 31$, $n \le 320$].

The mechanism, which we will need again in §3.1, is a two-line lemma about harmonic letters.

Lemma 2.3 (Paper A [1])**.** [PROVED]. *Let χ be a completely multiplicative arithmetic function ($\chi \equiv 1$ allowed), $r \ge 1$, $y \ge 0$, and $K_\chi^{(r)}(y) = \sum_{m \le y} \chi(m) m^{-r}$. Then, as an identity in $\mathbb{Q}$,*

$$p^r K_\chi^{(r)}(y) \;=\; \chi(p)\, K_\chi^{(r)}(\lfloor y/p \rfloor) \;+\; p^r T, \qquad T = \sum_{m \le y,\ p \nmid m} \chi(m) m^{-r} \in \mathbb{Z}_{(p)}.$$

Proof. Split the sum at $p \mid m$. The p-divisible part is

$$\sum_{j \le \lfloor y/p \rfloor} \chi(jp)(jp)^{-r} = \chi(p)\, p^{-r} \sum_{j \le \lfloor y/p \rfloor} \chi(j) j^{-r},$$

and the remaining terms are p-integral. □

The factor $\chi(p)$ in (LB_w^χ) is exactly the $\chi(p)$ produced by the p-divisible layer of the harmonic weight. It is invisible in the A-row: the unit-root congruence $A(ap+r) \equiv A(a)A(r) \pmod p$ is untwisted for all fifteen families [16, 26]. The character appears only on the non-unit Frobenius eigenvalue.

2.3 Existence of the tower limits

Theorem 2.4. [PROVED], *given* (LB_w^χ) *in ratio form at every level of the tower. Let* $p \ge 5$, $p \nmid a$, $n_s = ap^s$, *and suppose* $A(n_s)$ *is a* p*-unit cell for all* s, *so that the ratio form*

$$p^w f(n_s) \;\equiv\; \chi(p)\, f(n_{s-1}) \pmod{p^w}, \qquad f = B/A,$$

holds. Put $L_s := \chi(p)^s p^{ws} f(n_s)$. *Then*

$$L_s \equiv L_{s-1} \pmod{p^{ws}},$$

so $(L_s)_{s \ge 0}$ *is Cauchy in* $\mathbb{Q}_p$ *and* $\Lambda_a := \lim_{s\to\infty} L_s$ *exists, with* $\Lambda_a \equiv L_s \pmod{p^{w(s+1)}}$. *The same holds along any descent branch* $n_s = pn_{s-1} + r_s$.

Proof. $L_s - L_{s-1} = \chi(p)^s p^{w(s-1)}\big(p^w f(n_s) - \chi(p) f(n_{s-1})\big)$, and the bracket has valuation $\ge w$; hence $v_p(L_s - L_{s-1}) \ge ws$. The sequence is therefore Cauchy, and telescoping the tail gives $v_p(\Lambda_a - L_s) \ge w(s+1)$. Nothing in the argument used $n_s = ap^s$ beyond $n_{s-1} = \lfloor n_s/p \rfloor$. □

For the classical Apéry pair ($w = 3$, $\chi = 1$) this is

$$L_s = p^{3s} \frac{b_{ap^s}}{a_{ap^s}}, \qquad L_s \equiv L_{s-1} \pmod{p^{3s}}, \qquad \Lambda_a = \lim_s L_s.$$

For the Brown–Zudilin rows we write, in the same spirit,

$$\Lambda_a^P := \lim_s p^{5s} \frac{P_{ap^s}}{Q_{ap^s}}, \qquad \Lambda_a^{\widehat{P}} := \lim_s p^{3s} \frac{\widehat{P}_{ap^s}}{Q_{ap^s}}.$$

For the P-row Theorem 2.4 does *not* apply as stated, because the naive weight-5 ratio descent degrades; see §3.4.

2.4 What the limit does and does not remember

Proposition 2.5. [PROVED]. *In the situation of Theorem 2.4:*

(i) Λ_a *is unchanged by* $B \mapsto B + cA$ *for any* $c \in \mathbb{Q}$;

(ii) Λ_a *scales by* λ *under* $B \mapsto \lambda B$;

(iii) $\Lambda_{p^k a} = \chi(p)^k p^{-wk} \Lambda_a$ *for* $k \ge 0$ *(so* $\Lambda_{p^k a} = p^{-wk}\Lambda_a$ *whenever* χ *is trivial);*

(iv) $\Lambda_a \equiv f(a) \pmod{p^w}$*, provided* $A(a)$ *is a* p*-unit cell (which is a standing hypothesis of Theorem 2.4, and is not automatic: see Remark 2.6).*

Proof. (i) The perturbation contributes $\chi(p)^s p^{ws} c \to 0$. (ii) is immediate. (iii) Reindexing $t = s + k$ in $\Lambda_{p^k a} = \lim_s \chi(p)^s p^{ws} f(ap^{s+k})$ gives $\chi(p)^{-k} p^{-wk} \Lambda_a$, and $\chi(p)^{-k} = \chi(p)^k$ for a quadratic (or trivial) character at an unramified prime. (iv) is the case $s = 0$ of $\Lambda_a \equiv L_s \pmod{p^{w(s+1)}}$ in Theorem 2.4. □

Item (i) is the reason Λ_a is an intrinsic invariant of the recurrence and the tower rather than of the particular second solution — and equally the reason it cannot "see" the archimedean normalisation $\lim_n b_n/a_n = \zeta(3)$ except through the single global scalar. This is the first structural sign that the expectation $\Lambda \leftrightarrow \zeta_p(3)/6$ is misplaced; §5 makes it a theorem-strength negative and §6 explains it. Item (iii) is used throughout as an internal consistency check on the numerics. Item (iv) is the "trivial floor": on a p-unit cell, any ansatz expressing Λ_a through $f(a)$ reproduces w digits for free, so only agreement substantially beyond w digits is evidence of anything. This is the calibration used in §6.6.

Remark 2.6 (how sharp (iv) is, and where it stops). Both qualifications matter, and we record the sweep because the floor is a calibration the rest of the paper leans on. For the Apéry pair ($w = 3$) over the 44 cells $p \in \{5, 7, 11, 13\}$, $a \leq 12$, $p \nmid a$:

- 36 cells are p-unit cells, and (iv) holds at every one of them. It is *sharp*: $v_p(\Lambda_a - f(a)) = w = 3$ exactly at 26 of the 36, with 4 at $v_p = 4$, 2 at 5 and 1 at 6 — so nothing beyond w digits is free.
- The remaining 8 cells — $p = 5$, $a = 1, 3, 6, 7, 8, 9, 11$ and $p = 11$, $a = 5$ — have $v_p(A(a)) > 0$, and (iv) *fails* at every one of them: the difference-valuation is 1 at $p = 5$, $a = 1, 3, 7, 9$ and at $p = 11$, $a = 5$; -1 at $p = 5$, $a = 6, 8$; and -4 at $p = 5$, $a = 11$. These are exactly the cells excluded by Theorem 2.4's hypothesis.
- The natural strengthening to $p^{w+v_p(\Lambda_a)}$ is *false* even on p-unit cells: it fails at $p = 7$, $a = 3$ and at $p = 13$, $a = 2, 6, 10$, where $v_p(\Lambda_a) = 1$ so the claim would demand p^4 while the true difference-valuation is exactly 3.

The simplest counterexample needs no computation beyond Table 15: at $p = 5$ it prints $\Lambda_1 = 5^{-1}(1\,1\,3\,3\,4\ldots)$, while $f(1) = b_1/a_1 = 6/5 = 5^{-1}(1\,1\,0\,0\,0\ldots)$; they first differ in the third digit, so $v_5(\Lambda_1 - f(1)) = 1$, which is neither $\geq w = 3$ nor $\geq w + v_5(\Lambda_1) = 2$. (Here $a_1 = 5$, so $a = 1$ is not a 5-unit cell.)

3 The rate law

Theorem 2.4 gives a rate of at least w new p-adic digits per tower level. The measured rate is neither w nor constant; it is $\min(3, r(\chi, w))$, and both arguments of the minimum have a mechanism.

3.1 The harmonic tower and its rate

Let χ be a Dirichlet character (possibly trivial), $w \geq 2$, and set $K_\chi^{(w)}(y) = \sum_{m \leq y} \chi(m) m^{-w}$. By Lemma 2.3 the tower

$$h_p^{\chi,w}(a) \;:=\; \lim_{s\to\infty} \chi(p)^s p^{ws} K_\chi^{(w)}(ap^s)$$

exists, and its convergence rate is computable from the layer sums.

Finding 3.1. [VERIFIED: $p = 5, 7, 11$; $w = 2, 3$; $a \le 3$; $s \le 3$] The harmonic tower converges at the rates of Table 2.

Table 2: Rate of the χ-twisted harmonic tower, and the mechanism.

χ	w	rate $r(\chi, w)$	mechanism
1	2	3	$S_i \approx B_M p^i$; $\zeta_p(3) \neq 0$
1	3	5 (digits $4, 9, 14, 19$ at $p = 5$)	idem, but $\zeta_p(4) = 0$ kills the leading term
$\chi \neq 1$	2	2	$\sum \chi(k)k^{-w}$ tends to a *nonzero* L-value: no p^i gain
$\chi \neq 1$	3	3	idem

In words: for the trivial character the leading layer of the tower is divisible by an extra power of p, gaining one digit; for odd w it gains two, because the coefficient that would have carried the first correction is $\zeta_p(w + 1) = 0$. For a nontrivial character the layer sums converge to a nonzero generalised-Bernoulli/L-value and there is no gain at all. Thus

$$r(\chi, w) = w \ (\chi \neq 1), \qquad r(1, w) = \begin{cases} w + 1, & w \text{ even}, \\ w + 2, & w \text{ odd}. \end{cases}$$

[VERIFIED: $w = 2, 3$ only.]

3.2 The Jacobsthal–Kazandzidis cap

The second argument of the minimum is a cap, and it comes from binomial coefficients rather than from harmonic sums.

Finding 3.2. [VERIFIED: $p = 5, 7, 11, 13$; 12 pairs (A, B) per prime; $t \le 5$] The Jacobsthal–Kazandzidis tower $\binom{Ap^t}{Bp^t}$ converges at exactly 3 digits per level [19].

This is weight-independent. Since (§6) the non-harmonic part of Λ_a is assembled entirely out of the limits $c_p(A, B)$ of such towers, no tower limit built this way can converge faster than 3 digits per level, whatever its weight.

3.3 The rate law

Finding 3.3 (Rate law)**.** [VERIFIED: 15 sporadic families $\times$ $p \in \{5, 7, 11\}$, $N = 20\,000$, towers $a = 1, 2$] The number of new certified p-adic digits gained per level of the tower $n_s = ap^s$ is

$$\min\bigl(3,\ r(\chi, w)\bigr),$$

with $r(\chi, w)$ as in Table 2, in every cell except at (ε), (ζ), C, F, s_{18}, where blocks cancel. In particular the rate is 3 for all eight trivial-character families at both weights, and 2 for the χ-twisted weight-2 families B, C, E, F, s_{18}.

Remark 3.4 (the shape of the deviations)*.* They are *cumulative* excesses, not uniformly faster levels. At $p = 5$ the three families C, F and s_{18} behave identically — cumulative agreement $4, 5, 7, 9, 11, 13$ against the predicted $2, 4, 6, 8, 10, 12$ — so the excess is $+2$ acquired at the first level and then held, and the level-2 increment is 1, i.e. *below* the predicted 2. (B and E are on the nose at $p = 5$, and F is on the nose at $p = 7$, where instead C and s_{18} carry a $+1$.) At $p = 7$ the (ε) increments are $4, 5, 2, 3, 3$, one of which is again below the predicted 3. Everything not named in Finding 3.3 reproduces $\min(3, r(\chi, w))$ exactly.

Table 3 records the measurements.

Table 3: Digits gained per tower level, $a = 1$. "raw" is without the $\chi(p)^s$ twist; "tw." is with it. $\chi(p) = -1$ cells are the ones where the raw tower does not converge at all.

family	w	χ	$\chi(7)$	raw ($p = 7$)	tw. ($p = 7$)	tw. ($p = 11$)	tw. ($p = 5$)
A (Franel)	2	1	$+1$	3	3	3	3
D (Apéry $\zeta(2)$)	2	1	$+1$	3	3	3	3
s_7, s_{10}	2	1	$+1$	3	3	3	3
(α), (γ), (δ), (ε)	3	1	$+1$	3	3	3	3
B, C, F	2	χ_{-3}	$+1$	**2**	2	2	2 (twist needed)
s_{18}	2	χ_{-3}	$+1$	2	2	2	2 (twist needed)
E	2	χ_{-4}	$\mathbf{-1}$	**0**	**2**	2	2 ($\chi(5) = +1$)
(ζ)	3	χ_{-3}	$+1$	4	4	4	4 (twist needed)
(η)	3	χ_5	$\mathbf{-1}$	**0**	**3**	3	degenerate

Remark 3.5. A first pass over weight-3 and weight-5 data suggests the clean statement "the rate is always exactly 3, independently of the weight". That statement is an artefact of testing only trivial-character pairs, where $\min(3, r(1, w)) = 3$ for $w \geq 2$. Finding 3.3 is the corrected form. The moral is worth stating plainly: *the* 3 *is Jacobsthal–Kazandzidis, not the motivic weight.*

For the classical Apéry pair the rate is flat to a degree worth recording separately.

Finding 3.6. [VERIFIED: exact data to $n = 125\,000$: $p = 5$, $s \leq 7$; $p = 7$, $s \leq 6$; $p = 11, 13$, $s \leq 4$] For the Apéry $\zeta(3)$ pair the rate is 3 digits per level asymptotically, flat in p and in the tower base a; along a p-power tower $n_s = ap^s$ the valuation $v_p\big(p^{3s} f(n_s)\big)$ is eventually constant in s, so that the pole order of b_n/a_n along a tower is exactly $3\lfloor \log_p n \rfloor + O(1)$. See Table 4.

Remark 3.7 (two qualifications that the data forces). Neither "constant in s" nor "flat in the branch" can be strengthened to a universal statement.

- *The valuation can jump once, at the first level.* At $p = 5$, $a = 11$ we compute $v_p\big(5^{3s} f(11{\cdot}5^s)\big) = -4, -2, -2, -2$ for $s = 0, 1, 2, 3$, while $v_5(b_{11}/a_{11}) = -4$. So the constant is $v_p(\Lambda_a)$, which need *not* equal $v_p(b_a/a_a)$: here a_{11} is divisible by 5^4, so $a = 11$ is not a 5-unit cell, and Theorem 2.4 does not apply at $s = 0$. The jump is visible in the first entry of the corresponding agreement row, which is 0 rather than 3.
- *Along a general descent branch the rate is* 3 *only on average.* The last row of Table 4 ($p = 7$, branch $r = 1$) reads $4, 6, 9, 13, 15$, i.e. per-level increments $4, 2, 3, 4, 2$. It is the p-power towers, not all branches, on which the increment is 3 level by level.

Remark 3.8 (an honest precision budget). Three digits per level is a hard constraint on what can be computed. Sixty digits would require $s = 19$, i.e. $n = p^{19} \approx 10^{13}$ at $p = 5$; even the $O(n)$ closed-form-sum route, which needs only $K + O(\log n)$ digits of working precision rather than the $O(N^2)$ of the recurrence route, caps out near $n \approx 10^7$, i.e. about 33 digits. The certified

Table 4: Digits of agreement between L_s and L_{s-1} for the Apéry pair, after clearing the common $p^{\min v}$ shift.

p	tower	$v_p(\Lambda_a)$	agreement per level
5	$a = 1$	-1	2, 5, 8, 11, 14, 17, 20
5	$a = 2$	0	3, 6, 9, 12, 15, 18
7	$a = 1$	0	3, 6, 9, 12, 15, 18
7	$a = 3$	1	2, 5, 8, 11, 14
11	$a = 1$	0	3, 6, 9, 12
13	$a = 1$	0	3, 6, 9, 12
5	branch $n_s = p^{s+1} - 1$	1	5, 8, 11, 14, 17, 20
7	branch $r = 1$	0	4, 6, 9, 13, 15

precisions actually reached are in Table 5. What made the identification of §6 possible was therefore not more digits but exact closed forms, which can be evaluated to any precision at any p.

Table 5: Certified relative digits of Λ_a (Apéry pair).

p	N reached	$a = 1$	$a \le 4$	$a \le 12$
5	400 000	$s = 8$, **27**	24	21 ($a = 12$: $s = 6$)
7	250 000	$s = 6$, **21**	21–18	18
11	170 000	$s = 5$, **18**	15	12
13	380 000	$s = 5$, **18**	15	15

3.4 The weight-5 correction

The Brown–Zudilin P-row is the only weight-5 instance available to us, and it is the one where the naive statement fails.

Finding 3.9. [VERIFIED: exact ladders to $n = 5000$; $p \in \{5, 7, 11, 13\}$]

(i) $v_p\big(p^{5s} P_{ap^s}/Q_{ap^s}\big)$ is constant in s: weight 5 is the right and only normalising exponent for the P-row. Likewise $v_p\big(p^{3s}\widehat{P}_{ap^s}/Q_{ap^s}\big)$ is constant in s.

(ii) The rate is nevertheless 3, not 5, for both rows (Table 6).

(iii) The single-step ratio descent $p^5 f(n_s) \equiv f(n_{s-1})$ has depth $5 - 2s$, which *degrades* and goes negative at $s \ge 3$. The multi-digit statement that survives is the p^{5L}-scaled one,

$$p^{5s}\, P_{ap^s}\, Q_{ap^{s-1}} \;\equiv\; p^{5(s-1)}\, P_{ap^{s-1}}\, Q_{ap^s} \pmod{p^{3s}}, \tag{3.1}$$

with depth exactly $3s$.

(iv) The $\widehat{P}$-row, by contrast, has *constant* single-step depth 3, exactly like the Apéry b-row.

Equation (3.1) is the correct form of the "graded-weight" descent for the top row. It is worth emphasising what goes wrong with the naive form: it is not that the congruence is false at one level, but that its depth is $5 - 2s$, so that it is *true and useless* for $s \ge 3$. Only the Q-weighted, p^{5L}-scaled statement has a depth that grows with s, and its growth rate is $3s$ — once again the Kazandzidis 3, not the weight 5.

Table 6: Brown–Zudilin towers, $a = 1$: valuation of the normalised row and digits of agreement per level.

p	row (weight)	s range	$v_p(p^{ws} \cdot \text{row}/Q)$	digits of agreement
5	P (5)	0..5	$0, 0, 0, 0, 0, 0$	3, 6, 9, 12, 15
5	$\widehat{P}$ (3)	0..5	$0, 0, 0, 0, 0, 0$	3, 6, 9, 12, 15
7	P (5)	0..4	-1 (const)	2, 5, 8, 11
7	$\widehat{P}$ (3)	0..4	-1 (const)	2, 5, 8, 11
11	P (5)	0..3	0 (const)	4, 7, 10
13	P (5)	0..3	0 (const)	3, 6, 9

4 Valuation strata and the vanishing of the p-adic defect

4.1 The strata

Finding 4.1 (Strata)**.** [VERIFIED: $n \leq 5000$; $p \in \{5, 7, 11, 13, 17, 19\}$] Over *all* n, not merely along towers, and with $L = \lfloor \log_p n \rfloor$,

$$v_p(Q_n) \;\geq\; 0, \qquad v_p(\widehat{P}_n/Q_n) = -3L + O(1), \qquad v_p(P_n/Q_n) = -5L + O(1),$$

with the $O(1)$ bounded by 2 uniformly; explicitly the deviations lie in the sets of Table 7, all contained in $[-2, 1]$.

Table 7: Deviations of the row valuations from the predicted strata, all $n \leq 5000$.

p	$v_p(\widehat{P}_n/Q_n) + 3L$	$v_p(P_n/Q_n) + 5L$	$v_p(\widehat{P}_n/P_n) - 2L$
5	$\{-1, 0\}$	$\{0, 1\}$	$\{-1, 0\}$
7	$\{-2, -1, 0, 1\}$	$\{-2, -1, 0, 1\}$	$\{-2, -1, 0\}$
11	$\{-1, 0\}$	$\{-1, 0, 1\}$	$\{-1, 0\}$
13	$\{-1, 0, 1\}$	$\{0, 1\}$	$\{-2, -1, 0, 1\}$
17	$\{-1, 0\}$	$\{0, 1\}$	$\{-1, 0\}$
19	$\{-1, 0\}$	$\{-1, 0, 1\}$	$\{-2, -1, 0, 1\}$

Remark 4.2. $v_p(Q_n) = 0$ for every $n \leq 5000$ at $p = 5, 13, 17$, but reaches 7 at $p = 7$, 4 at $p = 11$ and 3 at $p = 19$: the exceptional-prime phenomenon is present in the Q-row and must be carried through any argument that divides by Q_n.

Finding 4.1 is a clean denominator statement about the whole Brown–Zudilin family, and it is precisely the input that a Rhin–Viola-style denominator saving [28] would have to improve upon.

4.2 $c_p = 0$

Finding 4.3 (the p-adic defect vanishes)**.** [VERIFIED: $n \leq 5000$; $p \in \{5, 7, 11, 13, 17, 19\}$] Since $\zeta_p(3), \zeta_p(5) \in p^{-1}\mathbb{Z}_p$ while the rows have poles of order $3L$ and $5L$,

$$v_p(\widehat{I}_n) - v_p(Q_n) = -3L + O(1), \qquad v_p(I'_n) - v_p(Q_n) = -5L + O(1),$$

hence $v_p(\widehat{I}_n/I'_n) = 2\lfloor \log_p n \rfloor + O(1) \to +\infty$ and

$$\boxed{c_p \;:=\; \lim_{n\to\infty} \frac{\widehat{I}_n}{I'_n} \;=\; 0 \quad \text{in } \mathbb{Q}_p.}$$

Finding 4.4 (the renormalised defect is not a constant)**.** [VERIFIED: $p = 5, 7, 11, 13$; $a = 1, 2$; 12–18 certified digits] Dividing out the p^{2s},

$$\widehat{c}_p(a) := \frac{\Lambda_a^{\widehat{P}}}{\Lambda_a^{P}} = \lim_s p^{-2s} \frac{\widehat{P}_{ap^s}}{P_{ap^s}}$$

is a well-defined element of $\mathbb{Q}_p$ with $v_p(\widehat{c}_p(a)) \in \{-2, -1, 0\}$; it is *not* rational (rational reconstruction sits at the noise floor) and it *depends on* a (different a agree to 0 digits). The renormalised p-adic defect is thus a function on the tower tree, not a constant.

4.3 The archimedean contrast: positivity versus grading

Archimedeanly the same two linear forms ride the *same* ray: $\log|I'_n|/n$ and $\log|\widehat{I}_n|/n$ both tend to $\lambda_2 = -2.47237\ldots$, so $c = \lim_n \widehat{I}_n/I'_n$ is a finite nonzero constant, namely $c = -1/(2\zeta(2)) = -3/\pi^2 = -0.30396\ldots$ [PROVED] in the companion [2]; and that constant is exactly what forces the Brown–Zudilin class off the minimal ray, because a positivity relation expressing I'_n as a positive combination of two further cell integrals pins the cellular class to the ray ρ_B, of which the impurity $\zeta(5) + 2\zeta(2)\zeta(3)$ of the top period is the visible shadow [2]. p-adically that degeneracy is broken: the Frobenius grading $\mathrm{diag}(1, p^3, p^5)$ puts Q, $\widehat{P}$, P into three *distinct* valuation strata 0, $-3L$, $-5L$ (Finding 4.1), so the two forms cannot ride a common ray and the defect collapses to $c_p = 0$ (Finding 4.3). The strategic asymmetry this exposes is worth stating once, plainly: *the p-adic side of this family is structurally clean but arithmetically weak, while the archimedean side is arithmetically strong but structurally obstructed*; there is no purity-defect obstruction p-adically at all, and what remains there is purely a size budget (§8 and Appendix B).

5 The exclusion theorem

We now come to the negative result, which we regard as the main arithmetic content of the paper. A natural and frequently voiced expectation is that, since $b_n/a_n \to \zeta(3)$, the p-adic tower limit Λ_a ought to be a $\zeta_p(3)$-avatar — "one ratio, two completions". It is not.

5.1 The reference values

We use $\zeta_p(s) = L_p(s, \omega^{1-s})$ as in §1.5, computed from a Volkenborn-integral series derived from scratch and validated on 980 independent interpolation identities; the construction, the proof that the series *is* L_p, and the full validation record are in Appendix A. We record here only the valuations, which matter for the strata argument:

$$v_p(\zeta_p(3)) = 0 \;\; (p \geq 5), \qquad v_p(\zeta_p(5)) = \begin{cases} -1, & p = 5, \\ 0, & p = 7, 11, 13, \end{cases}$$

the $p = 5$ case being the one where $\omega^{1-5} = \omega^{-4}$ is trivial because $(p-1) \mid 4$. [VERIFIED: $p \in \{5, 7, 11, 13\}$, 12–22 digits.]

5.2 No low-height $\mathbb{Q}$-affine relation

The search instrument is an exact-Fraction LLL with the residue column weighted by p^K, so that only genuine relations survive; it was validated first on planted relations and on true relations among Iwasawa logarithms ($\log_p 4 - 2\log_p 2 = 0$ at height 2; $\log_p 6 - \log_p 2 - \log_p 3 = 0$ at height 1; a planted $5x - 7\zeta_p(3) - 3 = 0$ at height 7) — [VERIFIED: recovered at every prime tested]. For an m-term relation certified to K digits the LLL noise floor is $p^{K/m}$; a genuine relation must appear *below* the floor.

Finding 5.1 (Exclusion)**.** [VERIFIED: $p = 5, 7, 11, 13$; $K =$ 12–27 certified digits; $a = 1, \ldots, 12$ with $p \nmid a$] No relation

$$c_0 + c_1\zeta_p(3) + c_2\Lambda = 0 \qquad \text{or} \qquad c_0 + c_1\zeta_p(5) + c_2\Lambda = 0$$

with integer height $\lesssim 10^4$ holds for Λ any of Λ_a (Apéry), Λ_a^P, $\Lambda_a^{\widehat{P}}$ (Brown–Zudilin). Every candidate returned sits *at* the noise floor, i.e. is nothing. The same is true of: the 2-term test $(\zeta_p(3), \Lambda_1)$; the normalisation-free test $\Lambda_a/\Lambda_1 \in \mathbb{Q}$; the tests against $\log_p 2$, $\log_p 3$, and the unit root of the weight-4 level-8 newform $\eta(2z)^4\eta(4z)^4$ that Beukers attaches to the Apéry numbers [5]; the 4-term test $(1, \zeta_p(3), \log_p 2, \Lambda)$; and algebraicity of degree ≤ 3. Tables 8 and 9 record the heights against the floors.

Table 8: Apéry pair: best LLL height for $(1, \zeta_p(3), \Lambda_1)$ against the 3-term noise floor $p^{K/3}$.

p	K (certified digits)	best height	noise floor $p^{K/3}$
5	23	1.0×10^5	2.3×10^5
7	21	4.9×10^5	8.2×10^5
11	15	4.4×10^4	1.6×10^5
13	15	1.9×10^5	3.7×10^5

Table 9: The wider hunt at $p = 5$, $K = 27$: best LLL height found for each (target, basis) pair. Nothing anywhere below the floor. $C_1 := \Lambda_1 - h_p(1)$.

target	$(1, \zeta_p(3), \cdot)$	$(1, \zeta_p(5), \cdot)$	$(1, \log_p 2, \cdot)$	$(1, \log_p 3, \cdot)$	$(1, \text{unit root}, \cdot)$	$(1, \zeta_p(3), \log_p 2, \cdot)$
Λ_1	1.2e6	1.3e6	1.5e6	9.6e5	8.0e5	3.5e4
$\widetilde{A}_1$	1.1e6	1.1e6	1.2e6	1.2e6	9.6e5	2.1e4
C_1	1.3e6	1.7e6	6.7e5	7.4e5	8.2e5	2.5e4
$h_p(1)$	6.3e5	1.0e6	1.0e6	6.8e5	1.2e6	4.8e4
floor	**2.0e6**	2.0e6	2.0e6	2.0e6	2.0e6	**5.2e4**

The Brown–Zudilin rows behave identically: best heights 6×10^3 to 1.4×10^4 for $(1, \zeta_p(5), \Lambda_a^P)$ and $(1, \zeta_p(3), \Lambda_a^{\widehat{P}})$, $a = 1, 2$, at $K =$ 12–18, against floors 1.5×10^4 to 1.7×10^4. Nothing below the floor. Also excluded: $h_p(1) \in \mathbb{Q}$ (height 5.2×10^{10} vs floor 6.8×10^{10}) and $h_p(1)$ algebraic of degree 2.

Remark 5.2 (why the margins are already decisive). Every Beukers–Vlasenko-style Frobenius constant — $-\zeta_p(3)/3$, $-8\zeta_p(3)/25$, $-35\zeta_p(3)/108$, $-\zeta_p(5)/5$, and their companions [7] — has height $\leq 10^2$. These are excluded by Finding 5.1 with a margin of two to four orders of magnitude. So the negative is not a close call for the candidates one would actually propose.

5.3 The sharp form of the negative

The LLL evidence bounds heights. There is a stronger and entirely structural statement which shows that no sharpening of estimates can help.

Finding 5.3 (polynomial divergence)**.** [VERIFIED: all $n \le 5000$; $p \in \{5, 7, 11, 13\}$]

$$v_p\big(b_n - \zeta_p(3)a_n\big) - v_p(a_n) = -3\lfloor \log_p n\rfloor + O(1),$$
$$v_p\big(Q_n\zeta_p(5) - P_n\big) - v_p(Q_n) = -5\lfloor \log_p n\rfloor + O(1).$$

Equivalently $|a_n\zeta_p(3) - b_n|_p \asymp n^3|a_n|_p$ and $|Q_n\zeta_p(5) - P_n|_p \asymp n^5|Q_n|_p$: the p-adic linear forms *diverge polynomially* instead of converging.

A Bel-type p-adic irrationality criterion [25, Lemma 4] requires $|a_n + b_n\xi|_p \le \exp(-\alpha n)$. Ours grows like a power of n. The gap is therefore infinite, not finite, and no sharpening of any estimate can rescue these pairs p-adically. Note the logical relation to Theorem 2.4: the normalisation by p^{ws} is precisely the act of dividing this divergence out, and what survives, Λ_a, is then no longer a linear form in anything.

5.4 The structural reason: order 3 is too short

Beukers–Vlasenko [7, Prop. 1.3] write the Frobenius structure of a MUM operator as $\mathcal{A}\big(y_i(t^p)\big) = p^i \sum_j \alpha_j y_{i-j}(t)$, so an operator of order 3 has only α_1 and α_2. Their Theorem 1.5 evaluates these:

$$\alpha_j = [x^j]\,\Gamma_p(x)e^{-\Gamma'_p(0)x} = [x^j]\exp\Big(-\sum_{m\ge 2}\zeta_p(m)\frac{x^m}{m}\Big). \tag{5.1}$$

Because there is no $\zeta_p(1)$ and $\zeta_p(2) = 0$, this gives

$$\alpha_1 = \alpha_2 = 0, \qquad \alpha_3 = -\tfrac{\zeta_p(3)}{3}, \quad \alpha_5 = -\tfrac{\zeta_p(5)}{5}, \quad \alpha_6 = \tfrac{\zeta_p(3)^2}{18}, \ \ldots$$

The Apéry $\zeta(3)$ operator $\theta^3 - t(34\theta^3 + 51\theta^2 + 27\theta + 5) + t^2(\theta+1)^3$ has order 3. Its p-adic Frobenius structure is therefore *trivial*: the available slots are literally zero, and $\zeta_p(3)$ first appears at α_3, which needs order ≥ 4. In particular Λ is definitely not an α_j. (That the Apéry operator is a symmetric square of an order-2 operator, leaving no room whatever, is [RECALLED–UNVERIFIED].)

5.5 Consistency with the literature

Finding 5.1 and Finding 5.3 are consistent with, and arguably explain, two facts in the literature. First, Calegari's p-adic irrationality theorem for $\zeta_p(3)$ at $p = 2, 3$ [11] does *not* use the Apéry pair: it uses overconvergent p-adic modular forms. Second, the successful p-adic constructions of Lai–Sprang–Zudilin [25] obtain their linear forms from a Volkenborn integral of a rational function of degree -3, exploiting two vanishing mechanisms unavailable archimedeanly: purity by degree ($\sum_k r_{n,1,k} = \lim_{t\to\infty} tR_n(t) = 0$, which kills the $\zeta_2(3)$ coefficient outright and leaves a genuine two-term form), and $\zeta_p(2k) = 0$, which kills the even-weight terms for free. Neither mechanism is available to the Apéry or Brown–Zudilin pairs, whose p-adic content, as we now show, is not a linear form at all but a Γ_p-value.

Finding 5.4. [VERIFIED: as in Findings 5.1 and 5.3] Within the range swept — $p \in \{5, 7, 11, 13\}$, relation heights $\lesssim 10^4$, $n \le 5000$ — there is no route to the irrationality of any $\zeta_p(2k+1)$ through the classical Apéry pair or the Brown–Zudilin pair.

We state the wider claim separately, as an expectation rather than a result, because it is the one place in this paper where we extrapolate past the sweep.

Remark 5.5. We expect the same at every prime, and Finding 5.3 is the reason: the obstruction found is not a height bound that a longer search might clear, but polynomial divergence of the linear forms, a structural property of the pair that does not depend on p. That is an argument, not a proof, and no numerical evidence outside the four primes above supports it.

6 Identification: what the tower limits are

The exclusions of §5 are not accidents of height. In this section we identify Λ_a completely: it is an explicit, infinite, convergent assembly of two ingredients, each of which has a closed form, and each of which involves *all* the odd p-adic zeta values at once.

6.1 The harmonic split and $h_p(a)$

Apéry's formula

$$b_n = \sum_k A(n,k)\big[H_3(n) + T(n,k)\big],$$

$$A(n,k) = \binom{n}{k}^2 \binom{n+k}{k}^2, \qquad T(n,k) = \sum_{m \le k} \frac{(-1)^{m-1}}{2m^3 \binom{n}{m}\binom{n+m}{m}},$$

splits $f = b/a$ into $H_3 + c_n/a_n$, where $H_3(n) = \sum_{k \le n} k^{-3}$. The harmonic part has an exact tower limit.

Theorem 6.1. [PROVED]. *For $p \nmid a$,*

$$p^{3s} H_3(ap^s) = \sum_i p^{3i} S_i(a), \qquad S_i(a) = \sum_{k \le ap^i,\ p \nmid k} k^{-3}$$

(the sum over i with $ap^i \ge 1$; over $0 \le i \le s$ when $1 \le a < p$), and consequently

$$h_p(a) := \lim_{s \to \infty} p^{3s} H_3(ap^s) = \sum_i p^{3i} S_i(a) = \sum_{\substack{x \in \mathbb{Z}[1/p] \\ 0 < x \le a}} x^{-3} \ \in \ \mathbb{Q}_p.$$

The series converges: $v_p\big(p^{3i} S_i(a)\big) \ge 3i$.

Proof. Splitting H_3 at $p \mid k$ gives $H_3(n) = p^{-3} H_3(\lfloor n/p \rfloor) + \sum_{k \le n,\, p \nmid k} k^{-3}$; iterating and multiplying by p^{3s} telescopes to the displayed identity. Re-indexing $x = k/p^i$ identifies the total with the sum over $x \in \mathbb{Z}[1/p]$, $0 < x \le a$, of x^{-3}: each such x is uniquely k/p^i with $p \nmid k$, and $v_p(x^{-3}) = 3i$, so the terms tend to 0 p-adically and the rearrangement is legitimate. □

[VERIFIED: $h_p(a)$ against $p^{3s} H_3(ap^s)$ for $a = 1, 2, 3,\ s \le 3,\ p \in \{5, 7, 11, 13\}$: agreement $4, 9, 14, 19$ digits.] Note that the harmonic tower converges at 5 digits per level at weight 3 — faster than the Apéry tower — exactly as Table 2 predicts. Only the terms with $i < K/4$ are needed for K digits, so h_p is computable to 40+ digits in milliseconds; this is what makes it usable as a reference constant in Table 9.

6.2 Kazandzidis limits in closed form

Definition 6.2. For integers $A > B \geq 1$ set $c_p(A,B) := \lim_{t\to\infty} \binom{Ap^t}{Bp^t} \in \mathbb{Q}_p$. The limit exists by Jacobsthal–Kazandzidis [19], at 3 digits per level (Finding 3.2).

Theorem 6.3. [PROVED] *(from Morita's factorial factorisation and [7, eq. (2)]), and* [VERIFIED: 12 pairs $(A,B) \times p \in \{5,7,11\}$, all digits matching, zero mismatches, at $K = 12$ digits except for the pair (p^2, p), which the run truncates to $K = 6$]. *Let $D = A - B$ and, for $l \geq 1$, $A_l = \lfloor A/p^l \rfloor$, $B_l = \lfloor B/p^l \rfloor$, $D_l = \lfloor D/p^l \rfloor$. Put*

$$T(A,B) := \prod_{l\geq 1} \frac{(-1)^{A_l+1}\Gamma_p(A_l+1)}{(-1)^{B_l+1}\Gamma_p(B_l+1)\cdot(-1)^{D_l+1}\Gamma_p(D_l+1)}$$

(a finite product, equal to 1 when $A < p$). Then

$$\begin{aligned} c_p(A,B) &= -\,p^{v_p\binom{A}{B}}\; T(A,B)\; \frac{\Gamma_p(A+1)}{\Gamma_p(B+1)\,\Gamma_p(D+1)}\; e^{\mathcal{Z}(A,B)}, \\ \mathcal{Z}(A,B) &= -\sum_{\substack{m\geq 3\\ m \text{ odd}}} \zeta_p(m)\big(A^m - B^m - D^m\big)\frac{p^m}{m(1-p^m)}. \end{aligned} \tag{6.1}$$

Proof sketch. Morita's factorisation [27] $n! = p^{v_p(n!)}\prod_{i\geq 0}(-1)^{n_i+1}\Gamma_p(n_i+1)$, $n_i = \lfloor n/p^i \rfloor$, applied to Ap^t, Bp^t, Dp^t turns $\binom{Ap^t}{Bp^t}$ into $p^{v_p\binom{A}{B}}$ times a ratio of Γ_p-values at the arguments $\lfloor Ap^t/p^i \rfloor + 1$ etc. The arguments with $i \leq t$ are $Ap^{t-i}+1$, and $\Gamma_p(1+x) = -\Gamma_p(x)$ on $p\mathbb{Z}_p$ moves them onto p-power points; the arguments with $i > t$ produce exactly the finite tail $T(A,B)$, which is independent of t. Applying [7, eq. (2)], $\log\Gamma_p(x) = \Gamma_p'(0)x - \sum_{m\geq 2}\zeta_p(m)x^m/m$, at $x = Ap^k$, Bp^k, Dp^k and summing the resulting geometric series over k gives the exponential factor; the $\Gamma_p'(0)$ terms cancel because $A = B + D$, and the even-m terms vanish because $\zeta_p(2k) = 0$. □

The verified pairs include cases with carries, with $p \mid A$, and with $v_p(c_p) = 1$: $(A,B) = (2,1), (3,1), (3,2), (4,1), (5,2), (7,3), (p,1), (p,2), (p+1,1), (p+1,p), (2p,p), (p^2,p)$.

The special case that drives the Apéry tower is worth displaying.

Corollary 6.4. [PROVED] *and* [VERIFIED: to *all* certified digits: 30 at $p = 5$ (from $n = 2\cdot 5^9$), 24 at $p = 7$, 21 at $p = 11$, 18 at $p = 13$]

$$c_p := \lim_{s\to\infty}\binom{2p^s}{p^s} = 2\exp\Big(-\sum_{\substack{m\geq 3\\ m \text{ odd}}}\zeta_p(m)\,\frac{(2^m-2)\,p^m}{m(1-p^m)}\Big).$$

Proof. Specialise (6.1) at $(A,B) = (2,1)$: $v_p\binom{2}{1} = 0$, $T(2,1) = 1$, and $\Gamma_p(3)/\Gamma_p(2)^2 = -2$. □

Remark 6.5 (the quantitative form of the exclusions). [VERIFIED: $p = 5, 7, 11, 13$] Truncating the series in Corollary 6.4 after the $\zeta_p(3)$ term reproduces only 4–5 correct digits out of 24–30. The constant genuinely needs the entire tower of odd p-adic zeta values. This is the mechanism behind Finding 5.1, quantified.

6.3 The structure theorem

Along $n = p^s$ one has $v_p\binom{p^s}{k} = s - v_p(k)$, so writing $k = jp^{s-i}$ with $p \nmid j$, $0 < j \le p^i$, the Apéry summand $A(n,k)$ has valuation $2i$ and

$$\lim_{s\to\infty} A\big(p^s, jp^{s-i}\big) = \big[c_p(p^i, j)\, c_p(p^i + j, j)\big]^2.$$

The inner Apéry weight at $m = j'p^{s-i'}$ contributes

$$\frac{p^{3s}}{2m^3\binom{n}{m}\binom{n+m}{m}} \quad\longrightarrow\quad \frac{(-1)^{j'-1}p^{3i'}}{2\,j'^3\, c_p(p^{i'}, j')\, c_p(p^{i'} + j', j')},$$

of valuation $2i'$. Assembling:

Finding 6.6 (Structure theorem)**.** [VERIFIED: $p = 5, 7, 11, 13$, block by block; each predicted block confirmed and the residual moving to exactly the next block] With $\widetilde{A}_a = \lim_s a_{ap^s}$ and $\widetilde{B}_a = \lim_s p^{3s} b_{ap^s}$ (so $\Lambda_a = \widetilde{B}_a/\widetilde{A}_a$),

$$\widetilde{A}_1 = 1 + \sum_{i\ge 0} \sum_{\substack{0<j\le p^i \\ p\nmid j}} \big[c_p(p^i, j) c_p(p^i + j, j)\big]^2, \tag{6.2}$$

$$\Lambda_1 = h_p(1) + \frac{1}{\widetilde{A}_1} \sum_{(i,j),(i',j')} \big(\text{block of valuation } 2(i + i')\big), \tag{6.3}$$

where the leading term of (6.2) is the $k = 0$ term of the sum defining a_n, the (i, j) block of (6.2) has valuation exactly $2i$, and the leading blocks of (6.3) are

$$(0,0):\ \frac{c_p(2,1)}{2}, \qquad (0,1):\ c_p(2,1)^2 \sum_{j=1}^{p-1} \frac{(-1)^{j-1}p^3}{2j^3\, c_p(p,j)\, c_p(p+j,j)} \ =:\ X_3.$$

Table 10 is the block-by-block verification: in each row the right-hand side is the truncation of the block sum and the entry is the valuation of the first discrepancy against the directly computed constant.

Table 10: Block-by-block verification of Finding 6.6.

check	first discrepancy at v_p	predicted
$\widetilde{A}_1 = 1 + c_p(2,1)^2$	3	≥ 2
$\widetilde{A}_1 = 1 + c_p(2,1)^2 + \sum_j [c_p(p,j)c_p(p+j,j)]^2$	6	≥ 4
$(\Lambda_1 - h_p(1))\widetilde{A}_1 = c_p(2,1)/2$	3	≥ 2
$(\Lambda_1 - h_p(1))\widetilde{A}_1 = c_p(2,1)/2 + X_3$	4 ($p = 5$), 5 ($p = 11, 13$), 6 ($p = 7$)	≥ 4

Remark 6.7. Every row confirms the predicted block, in the sense that the residual moves to the valuation of the next block. One row is worth flagging: for the last check the generating script predicted the first discrepancy at $v_p = 5$ and observed it at $v_p = 4$ at $p = 5$ (`struct_p5.txt`). The block decomposition predicts only ≥ 4 there, which is what the table records and what holds; but the sharper $v_p = 5$ that one might expect from the pattern of the other primes does *not* hold at $p = 5$.

Remark 6.8. The identification is complete in the sense that every ingredient of Λ_a is named and evaluated in closed form. What is *not* finite is the number of ingredients: each new level of the tower contributes genuinely new Kazandzidis limits $c_p(p^i, j)$. This is the precise sense in which Λ_a is "not a new constant" and simultaneously "not a known one".

6.4 Why every exclusion holds

Combining Theorem 6.3 with Finding 6.6: the tower limits live in the Γ_p-value ring. Each is a convergent $\mathbb{Q}$-combination of products of exponentials of infinite $\mathbb{Q}$-series in $\zeta_p(3), \zeta_p(5), \zeta_p(7), \ldots$ with p-power-weighted rational coefficients. No finite low-height $\mathbb{Q}$-relation involving $\zeta_p(3)$ alone — or $\zeta_p(5)$ alone — can hold, and Remark 6.5 quantifies how badly a $\zeta_p(3)$-only model fails (5 digits out of 24 for Λ_1; 4–5 out of 24–30 for c_p). The exclusion table is therefore not a collection of near-misses but the expected shadow of a structure theorem.

6.5 Positioning: values versus Taylor coefficients

Equation (5.1) and equation (6.1) involve the same function,

$$G(x) := \Gamma_p(x) e^{-\Gamma'_p(0)x} = \exp\Big(-\sum_{m\geq 2} \zeta_p(m) \frac{x^m}{m}\Big),$$

and they extract different information from it. Beukers–Vlasenko take the *Taylor coefficients of* G *at* 0: these are the Frobenius structure constants α_j, and for an order-3 operator only α_1, α_2 exist, both zero. Our tower limits are *values of* G at the p-adic points $x = Ap^k$, multiplied together over k and assembled over the tower tree. That is the precise sense in which the constants of this paper are p-adic Frobenius data, and simultaneously the precise reason they are not $\mathbb{Q}$-affine in any single $\zeta_p(m)$.

Three further points of orientation.

(1) *Terminology.* The Bloch–Vlasenko / Roy–Vlasenko *Frobenius constants* κ_n [29, 20] are archimedean periods (expansion coefficients of a generalised gamma function $\kappa(s) = s^2 \int_0^1 t^{s-1}\delta(t)\,dt$), conjecturally $\mathbb{Q}$-combinations of products of zeta values of total weight n. The Beukers–Vlasenko *Frobenius structure constants* α_j [7] are the p-adic objects of (5.1). Our Λ_a is the p-adic analogue of neither: it is the analogue of the Apéry limit itself, i.e. of a κ-type period.

(2) *Why the naive expectation fails.* The archimedean Apéry limit is a value at one singular point. The p-adic "limit" is a sum over the whole p-adic tree, and it therefore collects G at *every* level. The expectation $\Lambda \leftrightarrow \zeta_p(3)/6$ compares an object of the first kind with an object of the second.

(3) *Adjacent literature.* We also compared against the Frobenius-constant line of [29, 20] and the interpolated-Apéry-number and quasiperiod circle of [17]; the objects there are archimedean κ-type periods and q-expansion data, and none of them is our Λ.

(4) *Novelty.* We found no source naming these limits, and none stating the closed form of Corollary 6.4. The ingredients — Morita's factorisation, Jacobsthal–Kazandzidis, [7, eq. (2)] — are all classical, so (6.1) is presumably folklore-reachable; the derivation above is in any case self-contained and the formula is verified independently. [VERIFIED: as in Theorem 6.3.]

6.6 The one-generator refutation

It is natural to hope that the family $\{\Lambda_a\}_{p\nmid a}$ is generated over $\mathbb{Q}$ by a single constant ω_p — that the tower base a enters only through the elementary data (a_a, b_a). It does not.

Finding 6.9. [VERIFIED: $p = 5, 7, 11, 13$; $a \in \{1, \ldots, 12\}$, $p \nmid a$; 12–27 certified digits] Every ansatz in Table 11 fails, each at the measured trivial floor of 3 residual digits. On the p-unit cells that floor is exactly what Proposition 2.5(iv) hands over for free, and by Remark 2.6 no more than that is free; on the remaining cells it is an empirical observation only. A closing ansatz would show residual agreement ≈ available digits.

Table 11: The one-generator hunt. Entries are residual digits of agreement out of 12–18 available; the trivial floor is 3.

ansatz	result
$\Lambda_a = \alpha b_a + \beta a_a$	residual 0–1 — fails outright
$\Lambda_a = \alpha f(a) + \beta$	residual 3 — trivial floor only
$\Lambda_a = \text{const}$; $\Lambda_a = \alpha f(a)$	0 / 3 — fails
Möbius $(\alpha b_a + \beta a_a)/(\gamma b_a + \delta a_a)$	residual 0–2 — fails
Möbius in $(a_a, b_a, 1)$	residual 0–2 — fails
Möbius in the state vector $(a_a, b_a, a_{a+1}, b_{a+1})$	residual −1–0 — fails
$\widetilde{A}_a = \alpha a_a + \beta b_a$ (and variants)	residual 3 — trivial floor
$\widetilde{B}_a = \alpha a_a + \beta b_a$	residual 3–4 — trivial floor
$\widetilde{A}$ satisfies the Apéry recurrence?	residual $v_p = 3$ at every n — no
$\delta_a/\delta_b \in \mathbb{Q}$, $\delta_a := \Lambda_a - f(a)$	all reconstructions at the noise floor
$C_a/C_b \in \mathbb{Q}$, $C_a := \Lambda_a - h_p(a)$	all reconstructions at the noise floor
$(1, \Lambda_a, \Lambda_b)$ spans 2 over $\mathbb{Q}$	heights 9e4–6e5 vs floors 1–7e5 — noise
$\Lambda_1, \widetilde{A}_1, \widetilde{B}_1, C_1, h_p(1)$ algebraic of degree 2 or 3	all at the noise floor

Finding 6.6 explains why no ansatz can close. Λ_a is a sum over a two-index family of blocks of valuation $2(i + i')$, and each new level contributes genuinely new Kazandzidis limits. There is no finite-dimensional $\mathbb{Q}$-structure to find, hence no ω_p.

One classical-looking candidate deserves an explicit burial. The Casoratian of the Apéry pair is

$$a_{n+1}b_n - a_n b_{n+1} = -\frac{6}{(n+1)^3}, \tag{6.4}$$

[PROVED] from (1.1): $W_n := a_{n+1}b_n - a_n b_{n+1}$ satisfies $(n+1)^3 W_n = n^3 W_{n-1}$ with $W_0 = -6$. It is a pure p-power, $v_p(W_n) = -3v_p(n+1)$, so it carries *no* normalising information about a tower: it is invisible at $n = ap^s$ with $p \nmid a$. This is exactly why the Casoratian-corrected ansätze collapse onto the plain ones and fail with them.

Remark 6.10 (a methodological trap, recorded). The Fermat quotients $q_p(2), q_p(3), q_p(5)$, the Wolstenholme quotients H_{p-1}/p^2 and $H^{(2)}_{p-1}/p$, and B_{p-3} are *rational numbers* (e.g. $q_5(2) = 3$, $H_6/49 = 1/20$, $B_{10} = 5/66$). Placing them in an integer-relation search against 1 is vacuous: it "finds" relations such as $q_p(2) - 9 = 0$ and reports false candidates. They are congruence data modulo small powers of p, not independent p-adic constants. Similarly $\Gamma_p(1/2), \Gamma_p(1/3), \Gamma_p(1/4)$ are algebraic — $\Gamma_p(1/2)^2 = \pm 1$ by the reflection formula, and the others are Gauss-sum values by Gross–Koblitz [18] — so they are already covered by the algebraicity tests of Table 11. The Γ_p that matters here is Γ_p at p-power arguments, which is exactly what Theorem 6.3 produces.

7 Existence without archimedean limits

We close the positive part of the paper with what we regard as the cleanest single piece of evidence that the p-adic tower limit is an object about Frobenius rather than about archimedean asymptotics.

Three of the fifteen sporadic sequences have *no* archimedean Apéry limit. For (2.1) the characteristic roots solve $\lambda^2 - a\lambda + c = 0$ and for (2.2) they solve $\lambda^2 - 2a\lambda + c = 0$; the discriminants are

$$B: \ -27, \qquad (\delta): \ -128, \qquad (\eta): \ -16,$$

complex-conjugate pairs of equal modulus. Poincaré–Perron therefore supplies no dominant solution and $B(n)/A(n)$ does not converge — [VERIFIED: 0 digits of agreement between $n = 590$ and $n = 600$, against 380 digits for the convergent families]. Every other sporadic sequence has real distinct roots.

Finding 7.1. [VERIFIED: $N = 20\,000$; $a = 1, 2$; $p = 5, 7, 11$ for B and (δ), $p = 7, 11$ for (η)] All three of B, (δ), (η) have p-adic tower limits at every unramified $p \geq 5$:

- (δ) outright, at rate 3 (its character is trivial);
- B and (η) after the twist, i.e. for $\Lambda_a^\chi = \lim_s \chi(p)^s p^{ws} B(ap^s)/A(ap^s)$, at rates 2 and 3 respectively.

All three satisfy (LB_w^χ) with a flat floor, with characters χ_{-3}, trivial, χ_5 respectively. The one exclusion is the ramified prime: (η) has $\chi_5(5) = 0$ and no tower limit at $p = 5$, by Finding 7.2.

Finding 7.2 (the twist is necessary, and the ramified case degenerates)**.** [VERIFIED: $p = 5, 7, 11$; E and (η) at $p = 7$; B, C, F, (ζ), s_{18} at $p = 5, 11$] For a χ-twisted family at a prime with $\chi(p) = -1$ the untwisted tower does not converge at all (0 digits gained per level); it converges after normalising by $\chi(p)^{-s}$. This is the normalisation demanded by the twisted master law (LB_w^χ). At a prime dividing the conductor — (η) at $p = 5$, where $\chi_5(5) = 0$ — the tower degenerates entirely: $v_p(\Lambda)$ drifts and there is no convergence. This is the only ramified prime ≥ 5 among the fifteen families.

The ramified case is benign at the level of the congruence, if not of the limit: there (LB_w^χ) reads $p^w B_n A_q \equiv 0 \pmod{p^w}$, which is exactly the assertion that B_n is p-integral, and [VERIFIED: $n \leq 500$] $\min_n v_5(B_n) = 0$ — the pole simply does not develop at the ramified prime.

Finding 7.3 (no cross-family relations)**.** [VERIFIED: $p = 7$, all pairs of the fifteen families, 12–21 digits] Pairwise integer-relation searches for $(1, \Lambda_1^{\mathrm{fam}_1}, \Lambda_1^{\mathrm{fam}_2})$ find nothing below the noise floor. The tower limits of different families all lie in the same ring — the Γ_p-value ring of §6.4 — but are not $\mathbb{Q}$-linearly related to one another at accessible heights.

8 Open problems

Problem 8.1 (Name the constants)**.** Is $h_p(1) = \sum_{x \in \mathbb{Z}[1/p],\, 0<x\leq 1} x^{-3}$ irrational? Transcendental over $\mathbb{Q}$? We have excluded $h_p(1) \in \mathbb{Q}$ at height 6.8×10^{10} and algebraicity of degree 2 (§5.2), but nothing is proved. The same question for $c_p = \lim_s \binom{2p^s}{p^s}$, for which Corollary 6.4 gives a closed form entirely in terms of $\zeta_p(3), \zeta_p(5), \ldots$: is c_p transcendental? Is it in the field generated by the odd p-adic zeta values, and if so, is that containment strict?

Problem 8.2 (Upgrade the Findings)**.** Prove the multi-digit master descent (LB_w^χ) modulo p^w; Paper A [1] proves only the single-digit form modulo p. Prove the scaled weight-5 law (3.1) with its depth $3s$. Prove the strata bounds of Finding 4.1 with an explicit uniform $O(1)$, and deduce $c_p = 0$ (Finding 4.3) unconditionally.

Problem 8.3 (The rate law beyond $w = 3$)**.** Finding 3.1 is verified only at $w = 2, 3$. Prove $r(\chi, w) = w$ for $\chi \neq 1$ and $r(1, w) = w + 1$ (w even), $w + 2$ (w odd), and prove the cap: that a tower limit assembled from Kazandzidis limits cannot gain more than 3 digits per level. Explain the upward deviations at (ε), (ζ), C, s_{18}.

Problem 8.4 (Order ≥ 4: is there a ζ_p-carrying Apéry family?)**.** By §5.4 the value $\zeta_p(3)$ first occupies the Frobenius slot α_3, which requires an operator of order ≥ 4. Is there an Apéry-like family of order ≥ 4 whose tower limit *is* $\mathbb{Q}$-affine in $\zeta_p(3)$ — i.e. for which the naive "one ratio, two completions" expectation is true? Our exclusion is a statement about the order-3 and rank-3 families we can compute with, not about the phenomenon in general.

Problem 8.5 (The intrinsic meaning of the twist)**.** The factor $\chi(p)$ enters through Lemma 2.3, on the non-unit Frobenius eigenvalue only; it is invisible in the A-row. Any crystal-theoretic account of (LB_w^χ) must carry the information that for seven of the fifteen families the second Frobenius eigenvalue is $\chi(p)p^w$ and not p^w — i.e. that the family has a CM or quadratic-twisted modular parametrisation. What is the correct statement, and what happens at $p \mid \mathrm{cond}(\chi)$, where the tower degenerates (Finding 7.2)?

Problem 8.6 (The $O(1) \to \delta n$ bridge)**.** This is, in our view, the field's question, and we state it as such. No individual $\zeta_p(2k + 1)$ is known to be irrational for a single prime $p \geq 5$; the three known individual results all live at $p = 2, 3$. Fitting the same ledger shape to those three results (Appendix B) gives $\mathrm{margin}(p, w) = (w + 3) \log p/(p - 1) - w$, reproducing all three published irrationality measures exactly [CONJECTURAL FIT], and predicts the two nearest misses:

$$\zeta_5(3): \text{ deficit } 0.586, \qquad \zeta_3(5): \text{ deficit } 0.606.$$

Closing either requires a denominator saving of $e^{\delta n}$ with $\delta \approx 0.59$ resp. 0.61. The descent congruences (LB_w^χ) are the single-prime shadow of exactly this kind of saving — but they save $O(1)$ per prime (bounded depth, exactly w), hence $o(n)$ in total, where one needs δn. *Bridging $O(1) \to \delta n$ is the whole problem.* Appendix B shows, by exact reproduction of the Lai–Sprang–Zudilin ledger, that the d_n^5 budget there is tight: there is no free denominator saving hiding in the present construction, so the saving must come from a group action on a multi-parameter family [28] rather than from a sharper estimate on a fixed one. We record, as the complementary judgement, that we do *not* regard linear independence of $1, \zeta_p(3), \zeta_p(5)$ for a specific p as the promising target: two-term forms are already at the edge, and a three-term form would need the elimination technology of [15, 24], whose p-adic versions give asymptotic counts rather than a fixed small index set.

Problem 8.7 (Is the Γ_p-value ring the right home?)**.** Finding 7.3 says the tower limits of different families are not $\mathbb{Q}$-linearly related at accessible heights, while Finding 6.6 says they all lie in the ring generated by values of $G(x) = \Gamma_p(x)e^{-\Gamma'_p(0)x}$ at p-power arguments. Give that ring a structure theory. Is there a p-adic analogue of the weight filtration under which h_p, $c_p(A, B)$ and Λ_a have well-defined weights?

A The Kubota–Leopoldt implementation, and its validation

Every exclusion in §5 is only as good as the reference values $\zeta_p(3)$, $\zeta_p(5)$. We therefore built the Kubota–Leopoldt L-function from scratch, proved that what we built is L_p, and validated the implementation against an independently coded classical construction. This appendix is the credibility of the negative result.

A.1 The series

Start from $\zeta_p(s) = L_p(s, \omega^{1-s})$ [25, Def. 8], the Volkenborn integral $\int_{\mathbb{Z}_p} t^k \, dt = B_k$ with the convention $B_1 = -1/2$, and the expansion $\langle t + x\rangle^{1-s} = \langle x\rangle^{1-s}(1 + t/x)^{1-s}$ valid for $|x|_p > 1$. These give

$$\zeta_p(s, x) = \frac{1}{s-1}\langle x\rangle^{1-s} \sum_{k\geq 0} \binom{1-s}{k} B_k x^{-k},$$

and hence, with M any common multiple of q_p and $\mathrm{cond}(\chi)$,

$$L_p(s, \chi) = \frac{1}{(s-1)M} \sum_{\substack{0<j<M \\ p\nmid j}} \chi(j)\langle j\rangle^{1-s} \sum_{k\geq 0} \binom{1-s}{k} B_k \Big(\frac{M}{j}\Big)^k. \tag{A.1}$$

Convergence is exact and effective: $v_p(B_k) \geq -1$ by von Staudt–Clausen and $v_p\big((M/j)^k\big) = k\, v_p(M)$, so the k-th term has $v_p \geq k\, v_p(M) - 1$.

Proposition A.1. [PROVED]. *The right-hand side of* (A.1) *equals* $L_p(s, \chi)$ *for all* $s \in \mathbb{Z}_p \setminus \{1\}$.

Proof. Set $s = 1 - n$ with $n \geq 1$. The inner sum truncates, and using $B_n(x) = \sum_k \binom{n}{k} B_k x^{n-k}$ together with the classical $B_{n,\psi} = F^{n-1} \sum_{a=1}^{F} \psi(a) B_n(a/F)$, the formula collapses to

$$L_p(1-n, \chi) = -\big(1 - \chi\omega^{-n}(p)p^{n-1}\big)\frac{B_{n,\chi\omega^{-n}}}{n},$$

which is the defining Kubota–Leopoldt interpolation property [22], for *every* $n \geq 1$. The right-hand series of (A.1) is continuous in $s \in \mathbb{Z}_p$ (the convergence is uniform), and $\{1 - n : n \geq 1\}$ is dense in $\mathbb{Z}_p$, so the two agree on all of $\mathbb{Z}_p \setminus \{1\}$. □

A.2 Validation

Finding A.2 (980/980)**.** [VERIFIED: $p \in \{2, 3, 5, 7, 11, 13, 17, 19\}$; every character ω^e with $0 \leq e < p - 1$; every $n \in [2, 15]$; modulus p^{12}; 980 identities; zero mismatches] The interpolation identity $L_p(1-n, \chi) = -(1 - \chi\omega^{-n}(p)p^{n-1})B_{n,\chi\omega^{-n}}/n$ holds for our implementation against *independently coded* classical generalised Bernoulli numbers $B_{n,\psi}$: for trivial ψ, B_n with the Euler factor $1 - p^{n-1}$; for ψ of conductor p, $p^{n-1} \sum_{a=1}^{p-1} \psi(a) B_n(a/p)$.

Two further checks:

- *Independence of the auxiliary modulus.* [VERIFIED: $M = q_p \cdot m$, $m = 1, \ldots, 5$] identical values in every case. The formula (A.1) depends on a choice of M; the computed value does not.

- *A second, independent route.* The characterisation used by [25, 7] — $\zeta_p(s) = \lim \zeta(k)$ along negative integers $k \equiv s \pmod{p-1}$ with $k \to s$ p-adically, i.e. $-B_n/n$ at $n \equiv 1 - s \pmod{\operatorname{lcm}(p^N, p-1)}$ — gives [VERIFIED: 16/16 matches at $p \in \{5, 7, 11, 13\}$, $s \in \{3, 5\}$, $N \in \{2, 3\}$; e.g. $p = 13$, $s = 5$, $n = 26360$].

A.3 Values

Base-p digits, least significant first (Table 12). These are the inputs to Tables 8 and 9.

Table 12: $\zeta_p(3)$, base-p digits, least significant first.

p	digits of $\zeta_p(3)$
5	2 4 1 2 3 1 4 2 0 2 0 0 2 3 3 0 ...
7	1 3 5 4 1 0 2 0 2 4 1 2 5 0 4 6 ...
11	5 4 1 0 5 8 5 2 6 8 3 1 4 3
13	6 6 5 2 11 12 5 1 5 2 7 5 10 10

B The Lai–Sprang–Zudilin denominator ledger, reproduced

Problem 8.6 rests on the assertion that the denominator budget of the known p-adic construction is tight. We verified this by regenerating the construction of [25] exactly.

Their object is $R_n(t) = 2^{8n}(2t+n)(t+1/2)_n^4/(t)_{n+1}^4$, of degree -3, with $S_n = -\int_{\mathbb{Z}_2} R_n'(t + 1/2)\,dt$; the two vanishing mechanisms recalled in §5.5 reduce S_n to a two-term form $S_n = \rho_{n,0} + \rho_{n,3}\zeta_2(5)$. The integrality of $\rho_{n,3}$ comes from a very-well-poised ${}^9V_8$ differentiated once in ε, turned by Andrews' transformation [21] into the explicit double sum

$$\rho_n = \sum_{0 \le i \le k \le n} 2^{4(n-k)} \binom{2i}{i}^2 \binom{2n-2i}{n-i} \binom{2k-2i}{k-i} \binom{2k}{k}^2 \binom{2n-2k}{n-k}, \qquad \rho_{n,3} = 768\rho_n,$$

while the bound $v_p(\rho_{n,0}) \ge -5$ comes from the same machine at ${}^{13}V_{12}$ with three ε-derivatives — this is where the "-5" originates. The smallness exponent $\alpha = 16 \log 2$ comes from Sprang's Δ-operator [30] in the estimate $v_2(\int f) \ge \Delta(f) - 1$ of [23, Lemma 2.4], and the growth exponent $\beta = 8 \log 2 + 5$ from the double root 2^8 of the characteristic polynomial $\lambda^2 - 2^9\lambda + 2^{16}$ together with the denominator cost $\Phi_n^{-1} d_n^6 = e^{5n + o(n)}$.

Finding B.1. [VERIFIED: $n \le 1000$, exact rational arithmetic] Regenerating $\rho_{n,0}$, $\rho_{n,3}$, ρ_n from the recursion of [25]:

(i) $\rho_{n,3} = 768\rho_n$ and $\rho_n \in \mathbb{Z}$;

(ii) the exact Casoratian $\rho_{n,0}\rho_{n+1,3} - \rho_{n+1,0}\rho_{n,3} = 3 \cdot 2^{16n+18}/(n+1)^5 \ne 0$, [VERIFIED: $n \le 60$ by regenerating both sequences from the recursion];

(iii) the conjectured denominator condition $d_n^5 \rho_{n,0} \in \mathbb{Z}$ holds, and d_n^4 does *not* suffice at any tested n;

(iv) the per-prime ledger $e_p(n) := -v_p(\rho_{n,0})$ equals $5\lfloor \log_p n \rfloor$ exactly for almost every p, with isolated deficits of 1–3 at $p = 3, 13$;

(v) $\log(\operatorname{den}\rho_{n,0})/n \to 5$ from below, the gap shrinking from 0.138 at $n = 200$ to 0.042 at $n = 1000$.

Consequently the d_n^5 budget is tight: there is no free denominator saving hidden in the construction.

B.1 The margin ledger

The three published individual p-adic irrationality results have the same shape — smallness $\alpha = 2 \cdot (\text{growth})$, denominator cost equal to the weight w, and $\text{growth}_p = (w+3)\log p/(p-1)$ — giving

$$\operatorname{margin}(p, w) = \alpha - \beta = (w+3)\frac{\log p}{p-1} - w, \qquad \mu \le \frac{\alpha}{\alpha - \beta}.$$

Finding B.2. [VERIFIED: the fit reproduces all three published irrationality measures exactly] See Table 13.

Table 13: The ledger fit against the published measures.

known result	α	β	$\mu = \alpha/(\alpha-\beta)$	published
$\zeta_2(3)$ [6, 11]	$12\log 2$	$6\log 2 + 3$	$7.1773988991\ldots$	$7.177398\ldots$
$\zeta_3(3)$ [6, 11]	$6\log 3$	$3\log 3 + 3$	$22.2814479514\ldots$	$22.281447\ldots$
$\zeta_2(5)$ [25]	$16\log 2$	$8\log 2 + 5$	$20.3426517389\ldots$	$20.342651\ldots$

Extrapolating the same ledger ([CONJECTURAL FIT]: three data points, all at $p \in \{2, 3\}$) gives the margins of Table 14. The *conclusion* is robust independently of the fit, in that nothing individual is known for any $p \ge 5$.

Table 14: Extrapolated margins $(w+3)\log p/(p-1) - w$. Positive means a proof exists by this route.

p	margin, weight 3	margin, weight 5
2	+1.1589 (known)	+0.5452 (known)
3	+0.2958 (known)	−0.6056
5	−0.5858	−1.7811
7	−1.0541	−2.4055
11	−1.5613	—
13	−1.7175	—

C Numerical data

Table 15 records the tower limits and the derived constants in base-p digits, least significant first, computed from exact rationals. Certified precisions: Apéry 24 digits at $p = 5$ (from $n = 78\,125$; 27 from $n = 400\,000$), 21 at $p = 7$, 15–18 at $p = 11, 13$; Brown–Zudilin 18 at $p = 5$ (from $n = 3125$), 15 at $p = 7$ (from $n = 2401$), 12 at $p = 11, 13$.

The Apéry column of Table 15 was computed twice, by independent programs using different algorithms (an $O(N^2)$ exact-rational recurrence and an $O(N)$ mod-p^K pass), with identical results.

Table 15: Base-p digits, least significant first, 12 digits shown; every constant is normalised by p^{-v} where v is its valuation. Entries marked † have $v = -1$; all others have $v = 0$.

p	Λ_1 (Apéry)	Λ_1^P (BZ)	$\Lambda_1^{\widehat{P}}$ (BZ)	$\widehat{c}_p(1) = \Lambda_1^{\widehat{P}}/\Lambda_1^P$
5	1 1 3 3 4 0 1 1 1 0 2 2†	3 3 2 2 4 0 0 2 3 0 1 3	4 2 3 0 1 4 4 3 3 4 1 1	3 4 2 0 4 3 0 4 0 4 2 2
7	4 1 4 4 6 3 0 5 2 6 4 5	2 6 5 1 3 3 2 4 0 6 5 1†	2 5 4 5 1 6 3 6 2 0 1 0†	1 3 4 5 4 4 2 6 2 5 4 1
11	10 8 8 4 10 7 0 6 5 5 5 4	3 1 5 7 4 7 2 9 7 9 1 10	5 10 6 5 7 3 8 2 3 0 0 0	9 7 6 9 1 7 7 5 9 4 9 5
13	9 2 5 9 6 6 7 1 10 8 11 7	8 12 6 7 8 3 12 2 12 10 12 3	6 10 4 2 0 7 7 8 8 11 0 11	4 8 4 6 11 8 11 11 10 5 11 10

Table 16: The ingredients of the structure theorem, base-p digits, least significant first, each normalised by p^{-v}. Entries marked † have $v = -1$, entries marked ‡ have $v = +1$; all others have $v = 0$.

p	Λ_1	$\widetilde{A}_1$	$h_p(1)$	$c_p(2,1)$
5	1 1 3 3 4 0 1 1 1 0 2 2†	1 0 2 0 2 4 3 0 1 0 3 1‡	1 0 0 0 4 3 1 0 3 2 1 0	2 0 0 2 0 0 2 2 2 2 3 1
7	4 1 4 4 6 3 0 5 2 6 4 5	5 0 0 0 5 3 3 0 4 1 1 6	1 0 0 0 0 4 3 6 1 3 5 4	2 0 0 3 1 0 3 1 6 5 2 3
11	10 8 8 4 10 7 0 6 5 5 5 4	5 0 0 7 3 5 6 0 3 9 8 3	1 0 0 0 0 5 5 10 3 0 10 0	2 0 0 2 4 4 2 4 5 3 2 0
13	9 2 5 9 6 6 7 1 10 8 11 7	5 0 0 7 0 4 11 12 4 6 3 8	1 0 0 0 0 12 11 4 9 12 8 4	2 0 0 2 0 2 4 0 4 7 5 8

D Reproduction

All computations use exact integer or rational arithmetic; no floating point occurs at any stage, and every p-adic digit reported above is derived from an exact rational. The programs are organised as follows.

- *Sequence generation*: exact Apéry and sporadic pairs from the integer recurrence (1.2) and its analogues; exact extension of the Brown–Zudilin ladders $Q, P, \widehat{P}$ by the certified order-3 recurrence, with the recurrence residual checked to be exactly 0 at every $n \leq 357$ on every row.
- *p-adic arithmetic*: a $\mathbb{Q}_p$ class carrying an honest precision bound through every operation, so that no digit is ever reported that is not certified.
- *Tower data*: L_s, Λ_a, $\widetilde{A}_a$, $\widetilde{B}_a$ for all towers and branches, with the per-level agreement recorded (Tables 4, 6).
- *Kubota–Leopoldt*: (A.1) plus both cross-checks of Appendix A.2.
- *Integer relations*: an exact-Fraction LLL with weighted residue column, plus rational reconstruction, self-tested on planted relations before use.
- *Identification*: h_p from Theorem 6.1; $c_p(A, B)$ both directly (as a limit) and from (6.1); the block-by-block check of Table 10.

Acknowledgements

[TO BE SUPPLIED]

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